- eSchool News - http://www.eschoolnews.com -

A sub-$4,000 supercomputer? It's no game

Posted By eSchool News Staff On January 17, 2008 @ 7:00 am In IT Management,Top News | Comments Disabled

Supercomputing in education is a brand-new game thanks to a cluster of Sony PlayStation 3 (PS3) consoles and some creative thinking.

In 2005, when University of Massachusetts at Dartmouth physics professor Guarav Khanna first learned of Sony's plans to release a new open-platform PlayStation device with a high-power microprocessor, or a "Cell Broadband Engine," he had a feeling the new video-game console would hold great promise beyond gaming.

Now, Khanna's suspicions have paid off, as he conducts high-level scientific research on a supercomputer he built with his colleague Glen Volkema late last year, using a cluster of eight PS3 consoles running Linux.

Because of the unique features and easy accessibility of the PS3, Khanna believes the console has huge potential for use in settings other than research institutions–such as in classrooms and school computer labs.

According to Khanna's calculations, each PS3 cell processor–which he dubs a "supercomputer on a chip"–is equal to about 25 processors in a traditional desktop computer. The PS3 has a "lot more legs" than other processors, Khanna says–"a lot more potential."

The raw power of Khanna's PS3 supercomputer is perfect for highly calculation-intensive research, such as his own on small stars and black holes. But it's the machine's low cost that is the real story.

Whereas a mid-range desktop computer with decent processing power costs about $1,000, a single PS3 console sells for about $400, even though the production cost of the device runs around $800. As with other gaming consoles, the cost of the PS3 is subsidized, Khanna explains, so Sony can make money from the sale of video games.

"We're talking about saving a tremendous amount of money," Khanna says, referring to the use of multiple PS3 consoles to create a supercomputer.

Khanna's eight-PS3 supercomputer has tremendous implications for the research community, especially as the wait time for supercomputer use has skyrocketed in recent years.

Owing to heavy use of the world's limited supply of supercomputers, calculations that should only take an hour can "take two days to start," Khanna explains, "because calculations are queued up." But if schools and colleges are able to create their own low-cost machines, that problem soon could be eliminated.

The PS3's potential as an ed-tech tool extends beyond research, Khanna says. Because the device can double as a regular desktop computer running Linux, PS3s could, for instance, be used in school computer labs for regular classroom activities–including writing papers and surfing the net.

Implementing a school PS3 lab likely would require overcoming perceptional hurdles about the use of gaming consoles in education, Khanna says, which could be "quite a battle." But over time, he says, educators and school IT personnel could build an awareness of the device's educational benefits.

Beyond that, there are few technical issues to consider. Except for a keyboard, mouse, and monitor, Khanna says, the PS3 comes bundled with all the computing pieces you need.

"The PS3 does it well," he concludes.

Links:

PlayStation3 [1]

Guarav Khanna, Ph.D. [2]

- [3]
- [4]
- [5]
- [6]
- [7]
- [8]
- [9]
- [10]

Article printed from eSchool News: **http://www.eschoolnews.com**

URL to article: **http://www.eschoolnews.com/2008/01/17/a-sub-4-000-supercomputer-its-no-game/**

URLs in this post:

[1] PlayStation3: **http://www.us.playstation.com/PS3**
[2] Guarav Khanna, Ph.D.: **http://www.umassd.edu/engineering/phy/people/faculty/gkhanna/welcome.cfm**
[3] Image: **http://www.facebook.com/share.php?u=http%3A%2F%2Fwww.eschoolnews.com%2F2008%2F01%2F17%2Fa-sub-4-000-supercomputer-its-no-game%2F&t=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game**
[4] Image: **http://twitter.com/home?status=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game%20-%20http%3A%2F%2Fwww.eschoolnews.com%2F2008%2F01%2F17%2Fa-sub-4-000-supercomputer-its-no-game%2F%20@eschoolnews**
[5] Image: **http://delicious.com/post?url=http%3A%2F%2Fwww.eschoolnews.com%2F2008%2F01%2F17%2Fa-sub-4-000-supercomputer-its-no-game%2F&title=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game&notes=University+of+Massachusetts+at+Dartmouth+physics+professor+Guarav+Khan power+console+has+huge+potential+for+use+in+settings+other+than+research+institutions--such+as+in+classrooms+and+school+computer+labs.**
[6] Image: **http://www.linkedin.com/shareArticle?mini=true&url=http%3A%2F%2Fwww.eschoolnews.com%2F2008%2F01%2F17%2Fa-sub-4-000-supercomputer-its-no-game%2F&title=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game&source=eSchool+News+--+eSchool+News+covers+the+intersection+of+technology+and+innovation+in+education.&summary=University+of+Massachusetts+at+Dartmouth+physics+p power+console+has+huge+potential+for+use+in+settings+other+than+research+institutions--such+as+in+classrooms+and+school+computer+labs.**
[7] Image: **http://www.stumbleupon.com/submit?url=http%3A%2F%2Fwww.eschoolnews.com%2F2008%2F01%2F17%2Fa-sub-4-000-supercomputer-its-no-game%2F&title=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game**
[8] Image: **http://www.eschoolnews.com/2008/01/17/a-sub-4-000-supercomputer-its-no-game/**

[9] Image: **mailto:?to=&subject=A+sub-%244%2C000+supercomputer%3F+It%26%238217%3Bs+no+game&body=University+of+Massachusetts+at+Dartmouth+physics+professor+Guarav+Khann power+console+has+huge+potential+for+use+in+settings+other+than+research+institutions--such+as+in+classrooms+and+school+computer+labs.%20-%20http://www.eschoolnews.com/2008/01/17/a-sub-4-000-supercomputer-its-no-game/**
[10] Image: **http://www.eschoolnews.com/feed/rss/**